

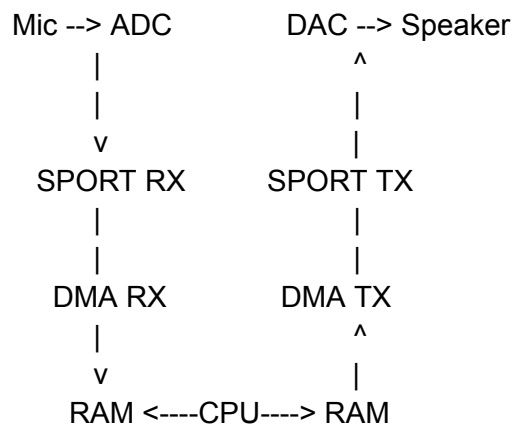
Delay Model

Ref mic → Direct Monitor → SHARC → Amp? → Acoustic Delay → Error mic

Direct Monitor

Sample rate	48kHz	96kHz
Delay samples	29	19
Delay ms	~0.6	~0.2

SHARC



$$D_{\text{total}} = D_{\text{ADC}} + D_{\text{CPU}} + D_{\text{DAC}}$$

<https://www.analog.com/media/en/technical-documentation/data-sheets/adau1761.pdf>

Table 5.

Parameter	Mode	Factor	Min	Typ	Max	Unit
ADC DECIMATION FILTER	All modes, typ @ 48 kHz					
Pass Band		$0.4375 f_s$		21		kHz
Pass-Band Ripple				± 0.015		dB
Transition Band		$0.5 f_s$		24		kHz
Stop Band		$0.5625 f_s$		27		kHz
Stop-Band Attenuation				67		dB
Group Delay		$22.9844/f_s$		479		μs

DAC INTERPOLATION FILTER

Table 5. (Continued)

Parameter	Mode	Factor	Min	Typ	Max	Unit
Group Delay	48 kHz mode, typ @ 48 kHz	$25/f_s$		521		μs
	96 kHz mode, typ @ 96 kHz	$11/f_s$		115		μs

$$D_{CPU} = 3 * \text{block_size}$$

$$D_{total} = D_{ADC} + D_{CPU} + D_{DAC}$$

$$D_{48kHz} = 23 + 3(4) + 25 + \sim 42\mu s \text{ (2 samples)} = 62 \text{ samples } (\sim 1.3 \text{ ms})$$

$$D_{96kHz} = 23 + 3(4) + 11 + \sim 42\mu s \text{ (4 samples)} = 50 \text{ samples } (\sim 0.52 \text{ ms})$$

The CPU ping pong procedure can be optimized if the callback is fast enough

Current procedure:

- Write previous output buffer to DAC buffer
- Read current ADC buffer to RAM
- Raise interrupt to run callback

Better procedure:

- Read current ADC buffer to RAM
- Directly call the callback
- Write current output buffer to DAC buffer

* This requires the callback to finish executing in 1 block size worth of time, or $\sim 41.7\mu s$ for 4 samples at 96kHz

$$\Rightarrow D_{CPU} = 2 * \text{block_size}$$

$$\Rightarrow D_{96kHz} = 46 \text{ samples } (\sim 0.48 \text{ ms})$$

Acoustic delay

$$\sim 4.5\text{cm} = \sim 12.6 \text{ samples @ } 96 \text{ kHz}$$

$D_{96\text{kHz}} = 19 + 46 + 13 = 78$ samples (~ 0.81 ms)

Recorded delay: 94 samples (~ 0.98 ms)

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>>> Relative delay: 94 samples
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>>> Relative delay: 0.979 ms
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